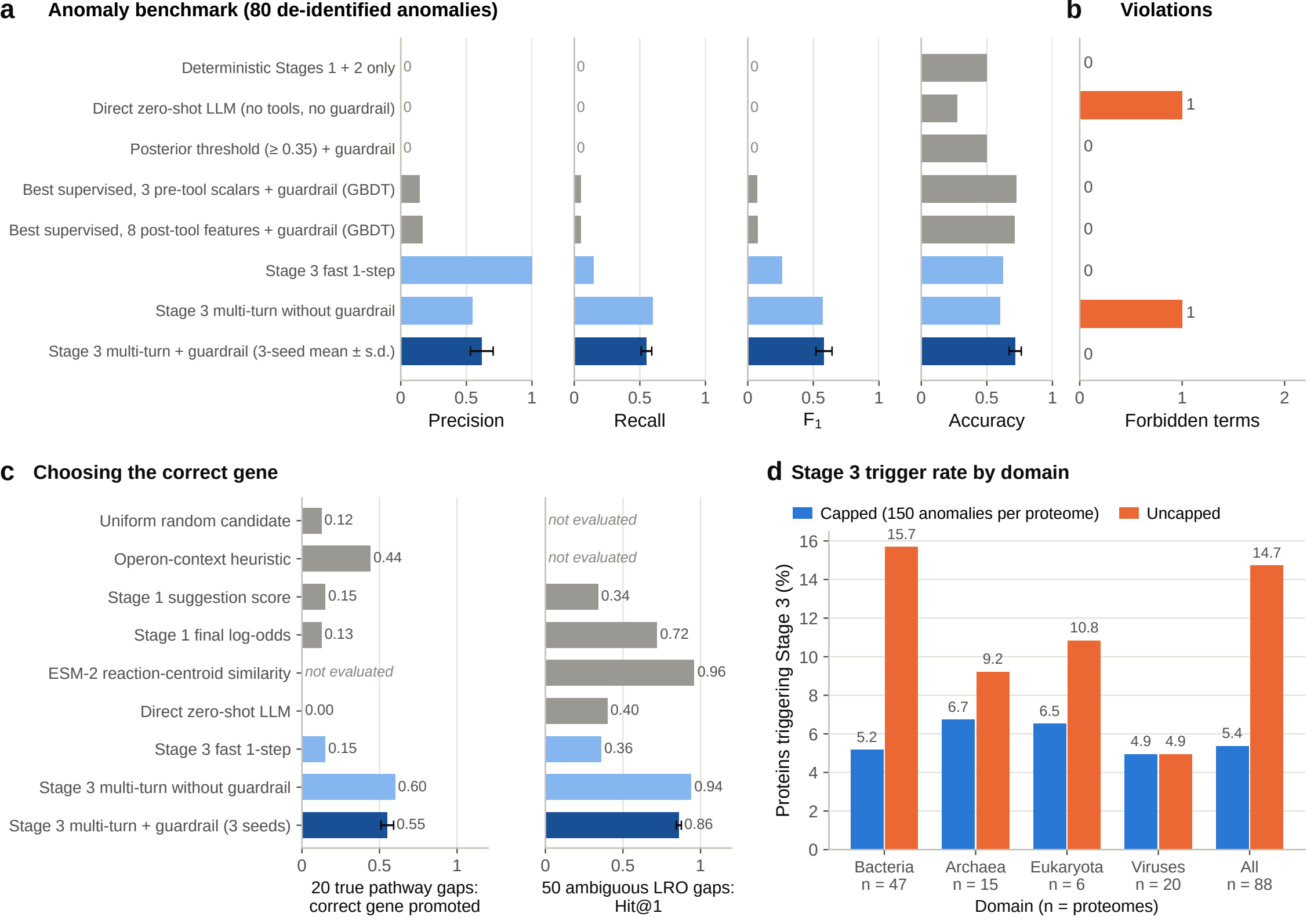
a
Anomaly benchmark (80 de-identified anomalies)
Deterministic Stages 1 + 2 only
Direct zero-shot LLM (no tools, no guardrail)
Posterior threshold (≥ 0.35) + guardrail
Best supervised, 3 pre-tool scalars + guardrail (GBDT)
Best supervised, 8 post-tool features + guardrail (GBDT)
Stage 3 fast 1-step
Stage 3 multi-turn without guardrail
Stage 3 multi-turn + guardrail (3-seed mean ± s.d.)
Precision
Recall
F1
Accuracy
b
Violations
Forbidden terms
c
Choosing the correct gene
Uniform random candidate
Operon-context heuristic
Stage 1 suggestion score
Stage 1 final log-odds
ESM-2 reaction-centroid similarity
Direct zero-shot LLM
Stage 3 fast 1-step
Stage 3 multi-turn without guardrail
Stage 3 multi-turn + guardrail (3 seeds)
not evaluated
20 true pathway gaps: correct gene promoted
50 ambiguous LRO gaps: Hit@1
d
Stage 3 trigger rate by domain
Capped (150 anomalies per proteome)
Uncapped
Proteins triggering Stage 3 (%)
Bacteria n = 47
Archaea n = 15
Eukaryota n = 6
Viruses n = 20
All n = 88
Domain (n = proteomes)